

Nonlinear stability of two magnetic puzzle-cube layouts

Root-mounted versus side-mounted magnets

Prepared for Jerich Lee

24 September 2026

Status. This is a numerical study of the explicitly selected baseline, not a measurement of the patented prototype. The root layout is inspired by the patent drawings; the side layout is an idealized comparator, not a specified commercial cube.

The main result is a tradeoff, not a universal winner

Both aligned equilibria are locally asymptotically stable in the reduced model. Root placement gives lower layer inertia, a wider direct attraction interval around alignment, and a slightly faster small-amplitude envelope decay at the same damping. However, the side layout has substantially greater local magnetic stiffness and **smaller first overshoot in the four matched release tests**.

Both assumed layouts also contain a shallow **unwanted stable equilibrium at 45°** . A release from 15° at rest returns to alignment in the root model but goes to the halfway equilibrium in the side model. A release from 30° goes to the halfway equilibrium in both. Thus local stability is not global return to the intended alignment.

Calculated quantity	Root-mounted	Side-mounted
Layer inertia, J (kg m ²)	1.0916×10^{-5}	1.1732×10^{-5}
Local magnetic stiffness, k (N m/rad)	0.441	1.510
Small-amplitude undamped frequency (Hz)	31.99	57.09
Damping ratio, ζ	0.0456	0.0238
Peak alignment-detent torque (mN m)	26.8	49.2
Nearest positive saddle angle	20.47°	11.84°
Aligned-well energy barrier (mJ)	4.637	4.685
First overshoot: $\theta_0 = 0$, $\dot{\theta}_0 = 5$ rad/s	1.340°	0.778°

What was actually computed

Finite-cylinder magnetic energies including all changing magnet pairs; nonlinear torque curves; equilibrium classification; free transients; energy-based capture bounds and sampled capture maps; conservative amplitude–frequency backbones; a true unforced return map; and a separately added periodic-torque experiment with stroboscopic maps, shooting, Floquet multipliers and frequency sweeps. Quadrature, interpolation and time-step refinements are included.

How to read the evidence

The patent supplies the internal edge–corner arrangement, not the numerical dimensions or damping used here.^[1] The course syllabus motivates phase-space, perturbation, Poincaré and Floquet tools.^[2] All numbers in this report are new calculations from the frozen baseline. The mathematical energy argument applies to the reduced model; numerical convergence checks are not experimental validation or interval-arithmetic certification.

1. Inputs and the mechanical question

The retained coordinate is the angle θ of the positive- z layer relative to the stationary remainder. The rotating assembly includes four complete edge pieces, four complete corner pieces and one center. All magnets attached to those pieces move with their owners, even when a magnet itself lies below an apparent face-layer boundary.

Input	Selected value, common to both layouts unless stated
Cube envelope	56 mm; vertical face-turn axis
Magnets	48 identical axial NdFeB cylinders; 3 mm diameter, 2 mm thickness
Magnetic prescription	$B_r = 1.20$ T; fixed uniform $M = B_r/\mu_0$
Magnet mass	Density 7500 kg/m ³ ; mass 0.1060 g each
Adhesive	0.005 g point mass at each magnet center
Aligned face gap	0.80 mm between facing magnet surfaces
Nonmagnetic masses	Edge 2.0 g; corner 2.5 g; center 2.0 g; fixed core/hardware 8.0 g
Mass distribution	Nonmagnetic centroids on an 18.667 mm lattice; each piece uses a 16 mm equivalent-cube centroidal inertia
Nominal dissipation	$c = 2.0 \times 10^{-4}$ N m s/rad, same in both models
Free-release input	No external torque after release

A controlled simplification

The turning layer is rigid. There is no independent cubie rocking, corner cutting, impact, axial spring displacement or body-to-body compliance. The fixed remainder is an imposed boundary condition. Gravity produces no generalized torque for the imposed vertical-axis rotation. No positive torsional spring is added to force a desired result.

Dry friction is **disabled** in this study. Values suggested in the input specification for possible static/kinetic friction are not applied. Likewise, suggested gap and placement-tolerance sweeps are not executed here. The only parameter sweeps reported are damping, diagnostic inertia swaps and the added forcing frequency.

How the magnet mass enters

For a rotating magnet with center offset ρ_i from the turning axis,

$$J_i = m_i \rho_i^2 + \hat{\mathbf{z}}^T \mathbf{I}_{i,G} \hat{\mathbf{z}}, \quad J = J_{\text{nonmagnetic}} + \sum_{i \in \text{moving}} (J_i + m_{\text{adh},i} \rho_i^2).$$

The disk inertia tensor is rotated with its axis. The computed total layer inertia falls by 6.95% in the root layout. This is not a 6.95% prediction of lower oscillation amplitude: the magnetic potential changes too.

Scope of “stability.” The analysis establishes restoring behavior for collective layer rotation. It cannot establish that each edge and corner is stable against its own admissible rocking or translation. Those are additional mechanical coordinates, not extra plotting options.

2. The actual magnet-coordinate models

Each marker below is a magnet center; each short arrow is its south-to-north direction. Both drawings use the same spatial scale. Filled markers move with the selected layer; hollow markers stay fixed. The box is only the nominal 56 mm envelope, not a collision-checked housing.

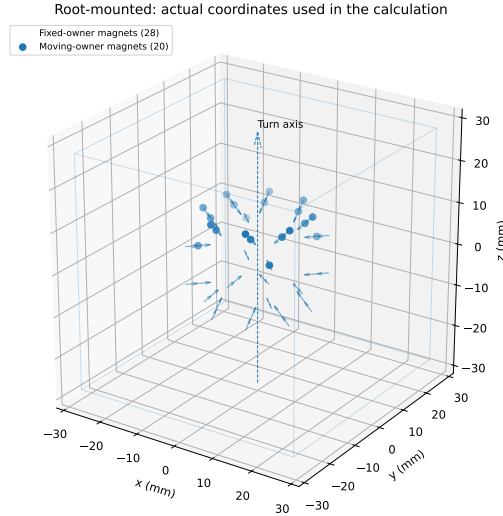


Figure 1: Root-mounted coordinate registry used in the field calculation. Three separate corner magnets are retained; they are not collapsed into one central dipole.

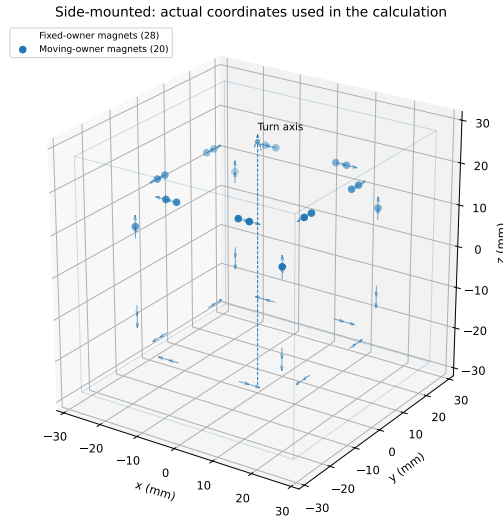


Figure 2: Side-mounted comparator. Motion follows piece ownership, not the magnet's own z coordinate.

The patent shows two edge-root insertion regions and three distributed regions around a corner base, with attractive edge-corner coupling (Figures 13–15, drawing sheets 10–12).^[1] The lengths and exact axis directions in these drawings of the *model* are selected assumptions. The full coordinate and polarity registries are supplied as CSV and JSON.

3. Finite-magnet interaction energy

For prescribed uniform magnetization, use the equivalent magnetic surface density $\sigma_m = \mathbf{M} \cdot \mathbf{n}$. An axial cylinder has $+M$ and $-M$ on its end disks and zero on its curved surface. This follows the standard magnetic-scalar-potential representation.^[3,4]

For two distinct, nonoverlapping magnets, the interaction energy used here is

$$U_{ij}(\theta) = \frac{\mu_0}{4\pi} \iint_{S_i \times S_j} \frac{\sigma_i(\mathbf{r}; \theta) \sigma_j(\mathbf{r}'; \theta)}{|\mathbf{r} - \mathbf{r}'|} dS dS'.$$

Each pair is counted once. Self-energies are constant. This is a finite-cylinder model with fixed M , not a point-dipole approximation and not a nonlinear ferromagnetic material solve.

All relevant interactions are retained

There are $\binom{48}{2} = 1128$ distinct pairs. Twenty magnets move and 28 remain fixed, giving $20 \times 28 = 560$ pairs with changing relative geometry. The remaining same-assembly pairs have constant energy under this coordinate. Their fields are not declared zero; their derivative with respect to the collective rotation is zero.

$$U(\theta) = \sum_{i \in \text{moving}} \sum_{j \in \text{fixed}} U_{ij}(\theta) + \text{constant}, \quad \tau_{\text{mag}}(\theta) = -U'(\theta).$$

The derivative includes both position and orientation changes. Consequently it includes force moment arms and direct magnetic torques, without adding them twice.

Numerical integration of the magnetic kernel

For a source-ring radius r , target cylindrical coordinates (ρ, z) and complete elliptic integral K , the azimuthal integral is

$$\int_0^{2\pi} \frac{d\phi}{\sqrt{\rho^2 + r^2 - 2\rho r \cos \phi + z^2}} = \frac{4K(m)}{\sqrt{(\rho + r)^2 + z^2}}, \quad m = \frac{4\rho r}{(\rho + r)^2 + z^2}.$$

The code uses SciPy's parameter convention for $K(m)$.^[5] Source radial integration uses 20 Gauss points. Each target end disk uses six radial Gauss points and 32 azimuthal points. A higher-order calculation and independent double-surface quadrature check the results.

Fourfold symmetry reduces the 20 moving magnets to five representatives, each weighted by four. Unreduced evaluations check that reduction. Reflection and 90° periodicity are also checked before extending the 0–45° grid at 0.1° spacing. A periodic cubic spline supplies *both* U and its torque derivative.

The solver is original code included in this package. Magpylib was **not** used to generate the numerical results. Finite-source modeling was retained because a point-dipole diagnostic overpredicts the local stiffness by approximately 143–145% at the selected small gap.

4. The full energy landscape has competing wells

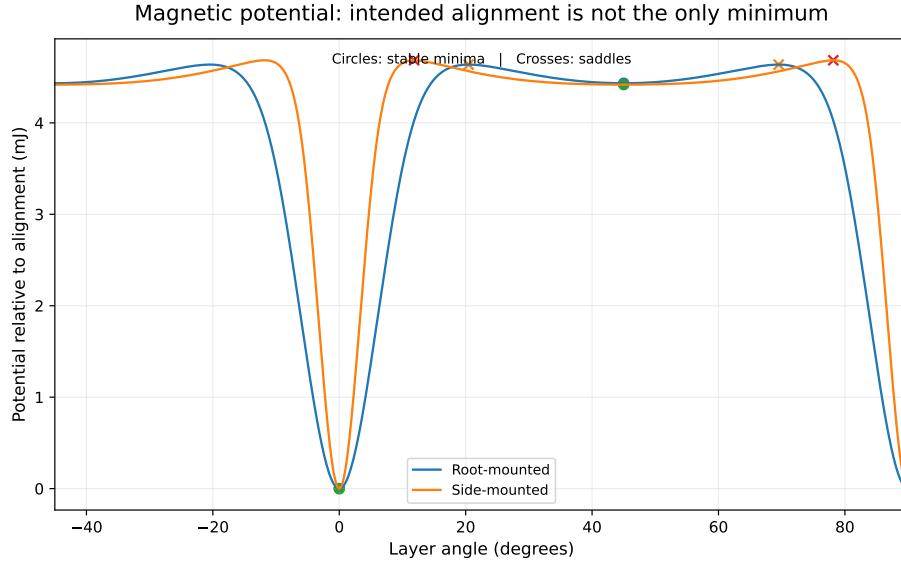


Figure 3: All changing magnetic interactions, with a separate zero-energy reference at alignment for each layout. Stable minima and saddle equilibria are not assumed in advance.

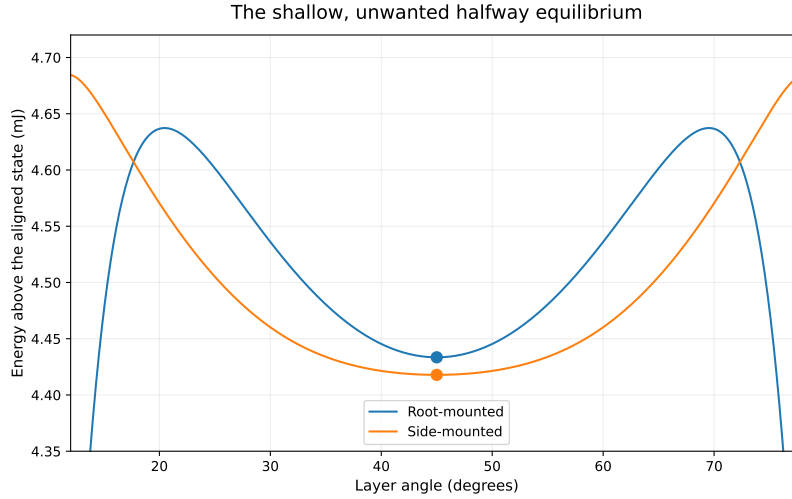


Figure 4: Magnification of the high-energy plateau reveals a shallow minimum at 45° in both models. The vertical scale is intentionally restricted to make that feature visible.

The nearest saddle lies at $\pm 20.47^\circ$ for the root layout and $\pm 11.84^\circ$ for the side layout. The halfway minimum is 4.433 mJ above alignment in the root case and 4.418 mJ in the side case. Its local depth is only 0.204 and 0.267 mJ, respectively, but a small depth does not remove the equilibrium in a frictionless-contact, viscously damped model.

These extra equilibria must not be hidden by substituting a single cosine or by fitting only a positive local spring. They are a consequence of the assumed complete magnetic geometry; their existence in the actual cube remains unverified.

5. Equation of motion and local stability

For each layout,

$$J\ddot{\theta} + c\dot{\theta} + U'(\theta) = \tau(t), \quad \dot{\mathbf{x}} = \begin{bmatrix} \omega \\ (\tau - c\omega - U'(\theta))/J \end{bmatrix}, \quad \mathbf{x} = (\theta, \omega).$$

The computed magnets remain separated under the imposed rotation; the finite-source potential is smooth. The periodic spline used in dynamics is C^2 , so its force is continuously differentiable. The state field is locally Lipschitz and the bounded periodic restoring torque with linear damping prevents finite-time blowup. Thus this reduced initial-value problem has unique continuing solutions for the considered bounded forcing.

At an equilibrium θ_* , $U'(\theta_*) = 0$. Linearization gives

$$\mathbf{A}_* = \begin{bmatrix} 0 & 1 \\ -U''(\theta_*)/J & -c/J \end{bmatrix}, \quad \lambda^2 + \frac{c}{J}\lambda + \frac{U''(\theta_*)}{J} = 0.$$

At intended alignment	Root	Side
$k = U''(0)$ (N m/rad)	0.4411	1.5098
Eigenvalues (s^{-1})	$-9.161 \pm 200.801i$	$-8.524 \pm 358.635i$
$f_n = \sqrt{k/J}/(2\pi)$ (Hz)	31.99	57.09
$\zeta = c/(2\sqrt{Jk})$	0.04557	0.02376
Envelope decay rate $c/(2J)$ (s^{-1})	9.161	8.524

Nonlinear energy argument

Around an isolated strict minimum define

$$V(\theta, \omega) = \frac{1}{2}J\omega^2 + U(\theta) - U(\theta_*).$$

For the released system ($\tau = 0$),

$$\dot{V} = \omega(J\dot{\omega} + U'(\theta)) = -c\omega^2 \leq 0.$$

In a sufficiently small compact invariant sublevel set, V is positive definite. The set $\dot{V} = 0$ has $\omega = 0$; remaining there also requires $U'(\theta) = 0$. If θ_* is its only equilibrium, the largest invariant subset is $(\theta_*, 0)$. LaSalle's principle gives **local asymptotic stability**.^[6]

Positive curvature at 0° is established numerically with refinement checks, and the argument applies to both reduced models. The same reasoning also makes the unwanted 45° minima locally attracting. Negative curvature at the intervening maxima produces saddles, not stable alignments.

The theorem is a model statement. No formally rounded interval proof of the magnetic integral, no full-cubic stability proof and no experimental stability certificate are claimed. Lyapunov/LaSalle is supplementary here: the supplied syllabus does not explicitly name it.

6. Restoring torque: lower inertia is not enough

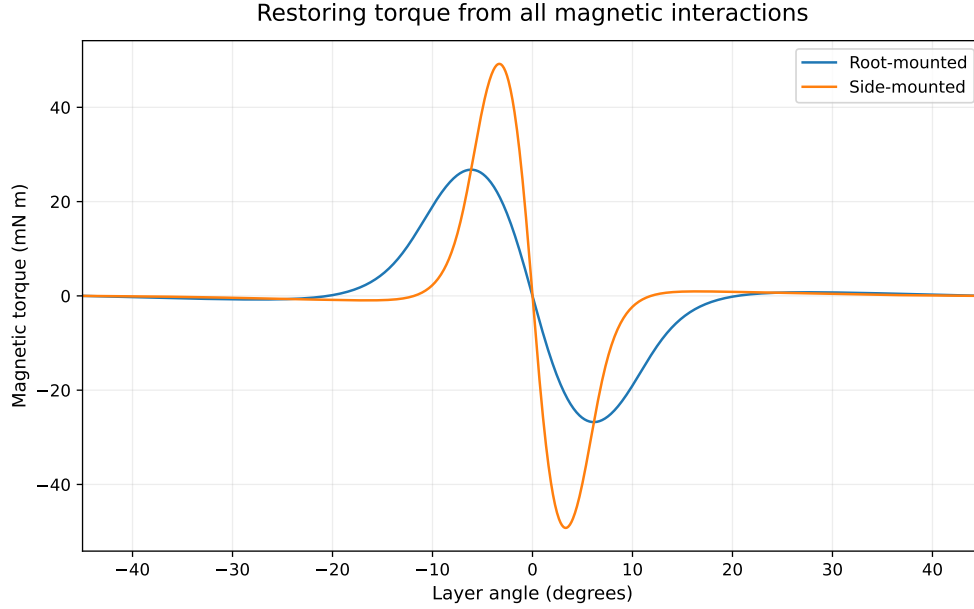


Figure 5: Torque is the negative derivative of the complete finite-cylinder potential. The root detent is broader and softer; the side detent is narrower and stronger.

The peak detent torque is 26.8 mN m at 6.13° for the root layout, compared with 49.2 mN m at 3.32° for the side layout. Lower turning torque is a different performance metric from low overshoot or a wide capture region.

Near alignment,

$$U'(\theta) = k\theta + \alpha\theta^3 + O(\theta^5),$$

with an even-potential fit over 0–1° giving

Coefficient	Root	Side
k from local series (N m/rad)	0.4410	1.5093
α (N m/rad ³)	−28.04	−340.61

The negative cubic coefficients imply local softening. The full magnetic potential, not this truncated polynomial, is used in all release and forced-response simulations.

For a small undamped same-speed release at alignment,

$$\frac{1}{2}J\omega_0^2 = \frac{1}{2}k\theta_{\max}^2 \quad \Rightarrow \quad \theta_{\max} \simeq |\omega_0|\sqrt{J/k}.$$

Root placement reduces J by about 7%, but in this geometry it reduces k by about 71%. The stiffness change outweighs the inertia benefit for this overshoot metric. That is the central reason the assumed inward placement does not automatically reduce every oscillation.

7. Two directly comparable free releases

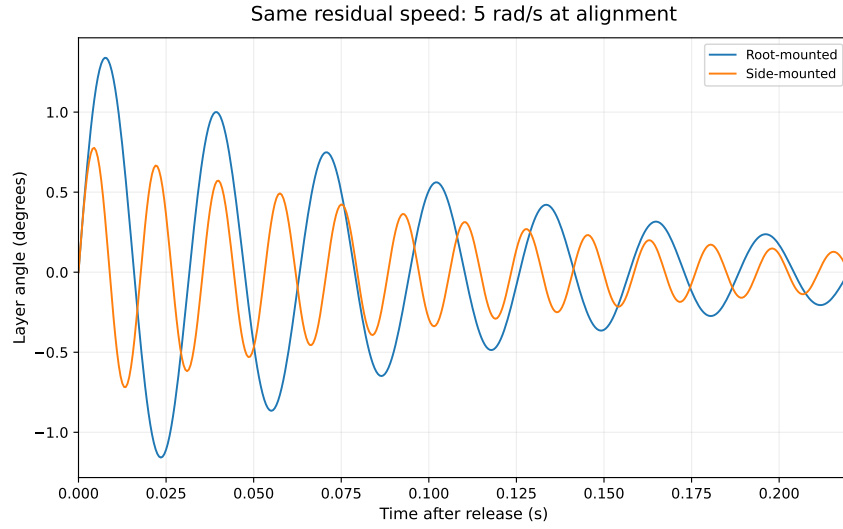


Figure 6: Same residual angular speed: both start aligned at 5 rad/s. Root first peak: 1.340° ; side: 0.778° . The plot shows the first 0.2 s; integration continues for 1 s.

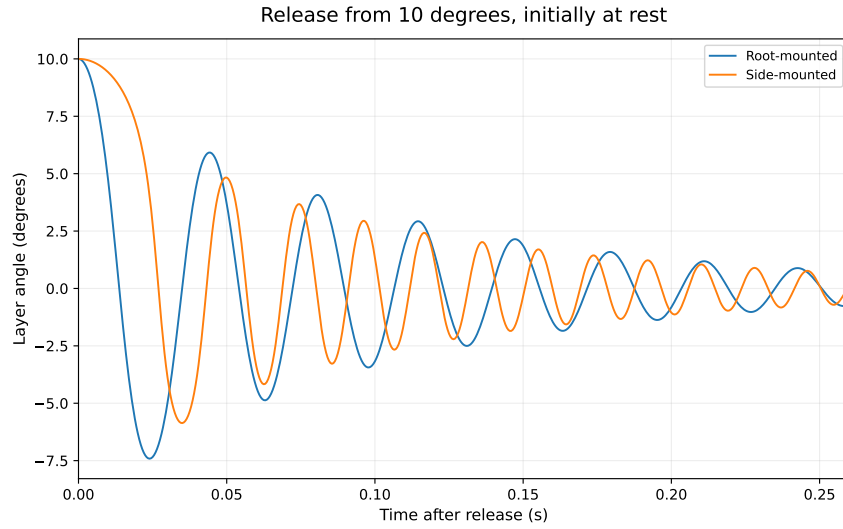


Figure 7: Same displacement: both start at $+10^\circ$ at rest. First opposite-side peak magnitudes are 7.414° and 5.865° . The two releases have different initial potential energies, because the potentials differ.

Both responses decay, but decay rate and first overshoot tell different stories. In the residual-speed test, the root response reaches the specified settling condition earlier (onset approximately 0.613 s versus 0.659 s) despite its greater initial excursion.

Settling definition: $|\theta| \leq 1^\circ$ and $|\omega| \leq 1^\circ/\text{s}$ continuously for 0.1 s. Confirmation occurs 0.1 s after the reported onset. For the 10° release, the observation is extended to 1.2 s: onsets are approximately 0.801 and 0.901 s. The side hold is completed just after 1 s, not absent forever.

8. Input matching changes the question

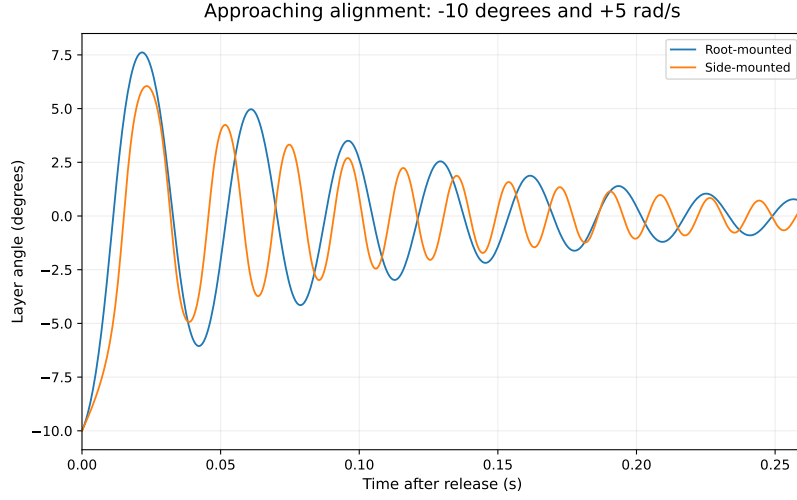


Figure 8: Both layers approach alignment from -10° with $+5$ rad/s. First positive overshoots are 7.617° for root and 6.052° for side.

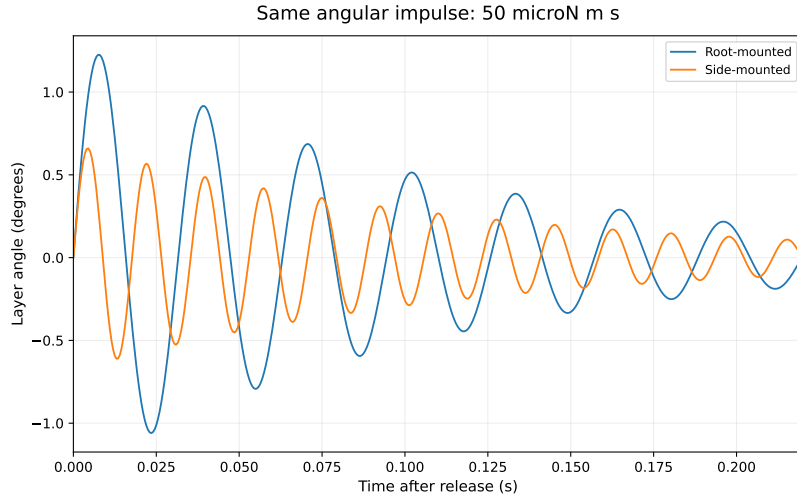


Figure 9: The same angular impulse $L = 5 \times 10^{-5}$ N m s is applied from rest, giving $\omega_0 = L/J$. Root starts at 4.580 rad/s and side at 4.262 rad/s; first peaks are 1.226° and 0.661° .

An impulse-matched comparison does not impose the same release speed. In the small conservative approximation,

$$\theta_{\max} \simeq \frac{|L|}{\sqrt{Jk}}.$$

Thus changing only inertia can have opposite effects under speed-matched versus impulse-matched tests. The four baseline release cases all give less first overshoot for the side layout, but none establishes a universal ordering over arbitrary hand inputs, damping, magnet geometry or coupled-piece motion.

The full trace and refined turning-point data are included in `data/free_metrics.json`. Peak angles are determined from velocity-zero events, not merely from a sparse maximum of sampled frames.

9. Phase portraits: aligned and unintended attractors

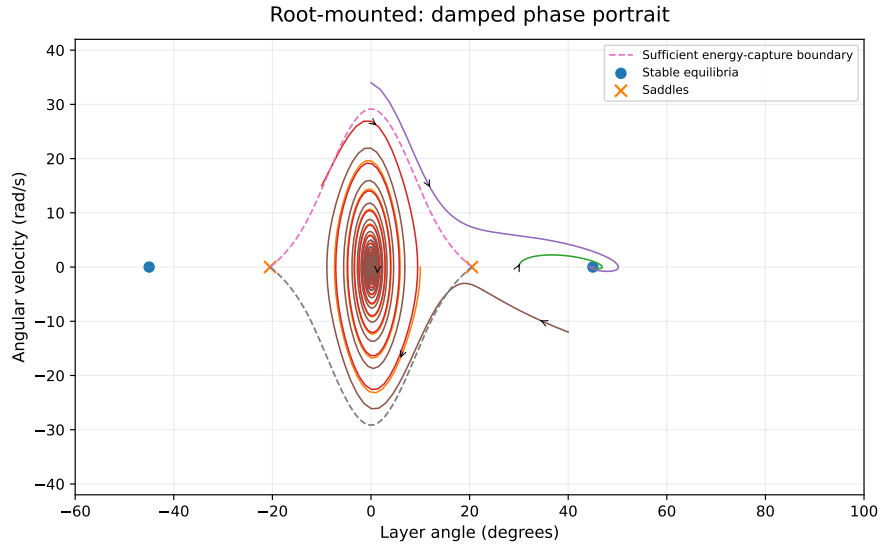


Figure 10: Root model: each continuous curve is a solution in (θ, ω) space. The curves around zero spiral inward; other initial states approach a halfway equilibrium. The dashed energy contour is a sufficient capture bound, not the exact damped separatrix.

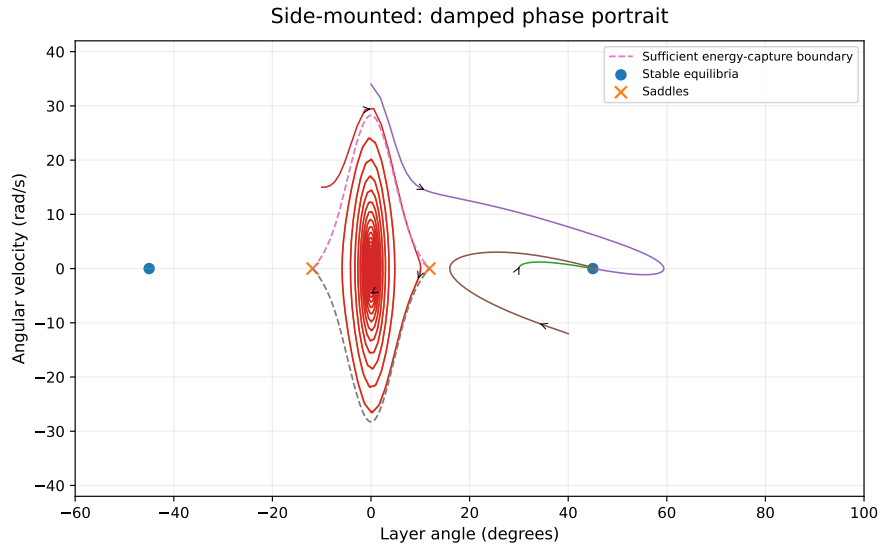


Figure 11: Side model, using the same angular and velocity scales. The intended-well attraction interval is narrower. Stable equilibrium markers and saddle markers identify the local phase-space structure.

Repeated loops do not imply a sustained limit cycle: their energy decreases. For any nonconstant periodic free orbit, integrating $\dot{V} = -c\omega^2$ over one period would require $\int c\omega^2 dt = 0$, a contradiction. No nonconstant periodic free orbit or sustained chaotic attractor is generated by this strictly dissipative one-coordinate model.

10. Capture maps answer a practical release question

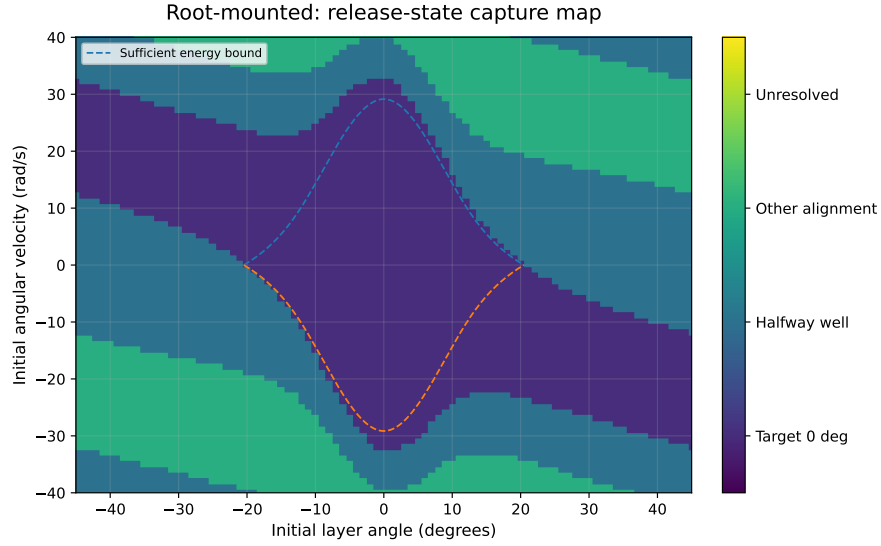


Figure 12: Root model: 7,371 initial conditions on a 91×81 grid, integrated for 3 s. Categories distinguish target alignment, halfway wells, other quarter-turn alignments and unresolved cases.

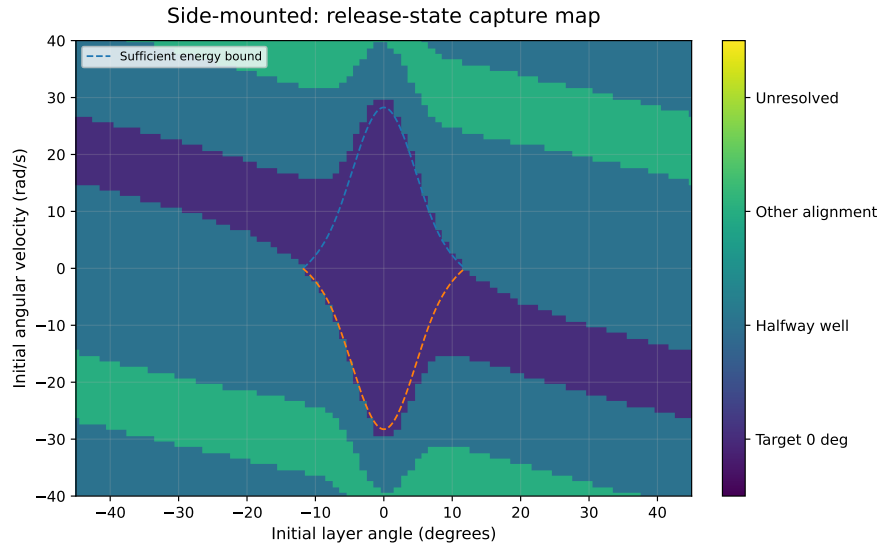


Figure 13: Side model, same initial-condition grid and integration settings. The energy-bound overlay is sufficient, not necessary: damping can capture trajectories that begin above the conservative barrier.

Initial angles span $[-45^\circ, 45^\circ]$ and initial speeds span $[-40, 40]$ rad/s. Final categories use a finite observation window and explicit tolerances. These colored areas are not statistical probabilities of a user's turns. A time-step-halving check at 837 grid points changes no category. The coarse grid does not constitute an exhaustive basin-boundary proof.

11. Energy bounds and a decisive capture example

Let θ_b be the first positive saddle and $\Delta U = U(\theta_b) - U(0)$. For an initial angle inside the intended well, the sufficient condition

$$|\theta_0| < \theta_b, \quad \frac{1}{2}J\omega_0^2 + U(\theta_0) - U(0) < \Delta U$$

keeps the trajectory below either barrier because energy cannot increase. The connected sublevel component contains only the aligned equilibrium, so LaSalle gives convergence to that equilibrium.^[6]

Bound quantity	Root	Side
θ_b	20.47°	11.84°
ΔU (mJ)	4.637	4.685
$\sqrt{2\Delta U/J}$ at alignment (rad/s)	29.15	28.26

The speed bounds are sufficient limits for the energy argument, not maximum capturable speeds after including damping. Likewise, θ_b is the zero-speed direct-attraction boundary, not the whole velocity-dependent basin.

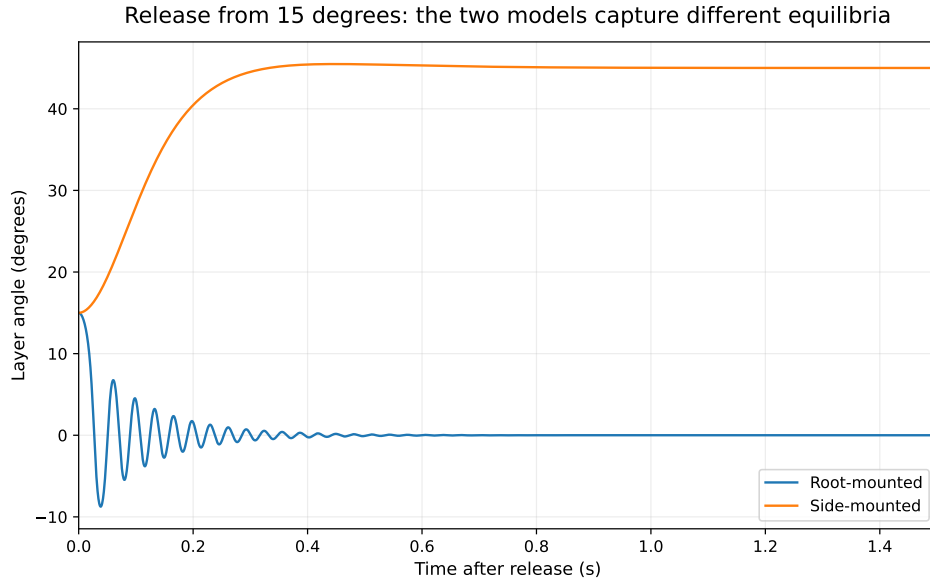


Figure 14: Identical 15°, zero-speed releases: root approaches 0°; side approaches 45°. This initial angle lies between the two calculated saddle locations.

Interpretation. The root geometry has a genuine advantage for this release condition even though it had larger small-angle overshoot in the earlier tests. Both findings can be true because capture destination and first overshoot measure different aspects of the dynamics.

12. Energy loss versus oscillation amplitude

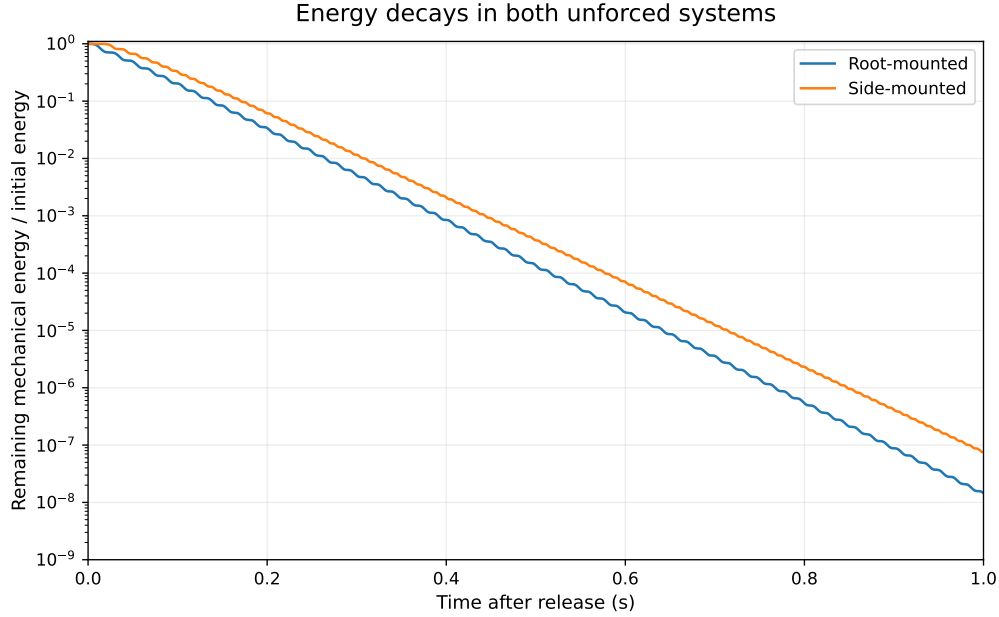


Figure 15: Total mechanical energy relative to alignment, normalized separately by each initial energy, for the 10° release. All intended and other magnetic interactions are already included in U .

The exact reduced-model balance after release is

$$E(t) + \int_0^t c\omega(s)^2 ds = E(0), \quad E = \frac{1}{2}J\omega^2 + U(\theta) - U(0).$$

An augmented integration of the dissipated-energy state verifies the balance with relative residuals of approximately 2.2×10^{-8} (root) and 3.7×10^{-8} (side) in the checked case. A coarse trapezoidal integral of saved output samples is a less accurate diagnostic and is not substituted for that check.

At sufficiently small amplitudes, the displacement envelope decays at rate

$$\sigma = \frac{c}{2J}, \quad A(t) \propto e^{-\sigma t}.$$

The rates are 9.161 and 8.524 s^{-1} . At the same assumed c , the root rate is about 7.5% greater. This is consistent with somewhat earlier settling, but does not require the root to have the smaller first excursion.

Magnetic stabilization is not automatic magnetic damping

The permanent-magnet energy here is conservative. With $c = 0$, energy is constant and the stable small oscillations do not decay. In this model the damping comes from the selected viscous term, not from the magnetic attraction itself. Eddy currents, hysteresis and contact friction are not separately resolved.

A static-friction extension would change the convergence target: resting states can satisfy $|U'(\theta)| \leq \tau_s$ rather than $U'(\theta) = 0$. That extension is not included in the reported results.

13. Nonlinear backbone and Lindstedt–Poincaré

For a conservative oscillation about alignment, set $c = 0$ only for this diagnostic. With turning amplitude $0 < A < \theta_b$, conservation of energy gives the period

$$T(A) = 4\sqrt{\frac{J}{2}} \int_0^A \frac{dq}{\sqrt{U(A) - U(q)}}, \quad f(A) = 1/T(A).$$

The numerical integral uses $q = A \sin \phi$ to remove the square-root endpoint singularity. It retains the complete magnetic energy, rather than treating stiffness as constant.

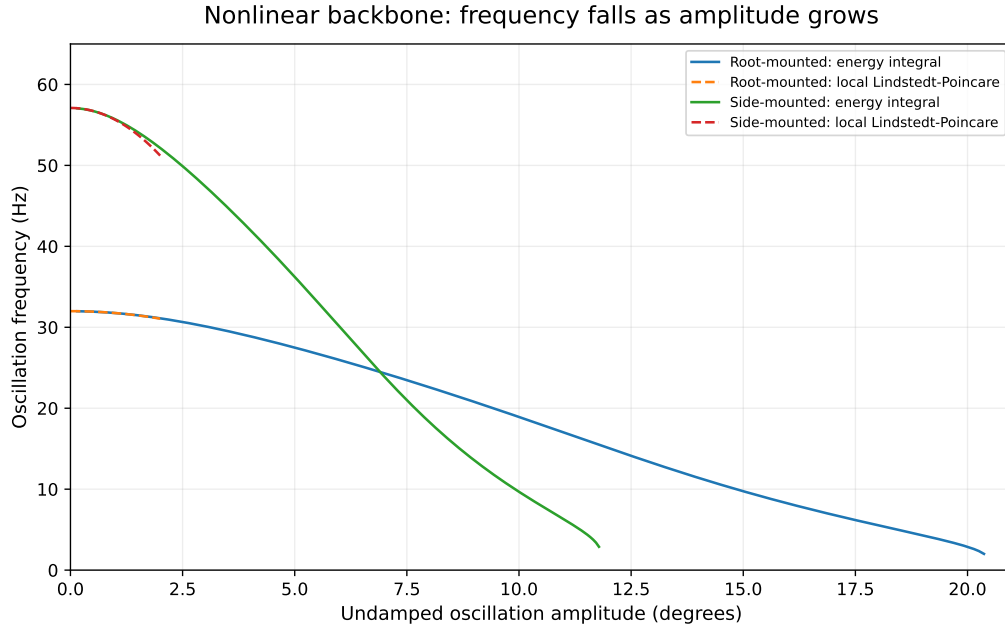


Figure 16: Both backbones soften: oscillation frequency falls as amplitude grows. Each full-energy curve stops before its own saddle. Dashed local asymptotic approximations are shown only over a small angular interval.

For $J\ddot{\theta} + k\theta + \alpha\theta^3 = 0$, the leading Lindstedt–Poincaré frequency correction is

$$\Omega(A) = \sqrt{k/J} \left[1 + \frac{3\alpha A^2}{8k} + O(A^4) \right].$$

The fitted $\alpha < 0$ in both layouts predicts the observed initial softening. The method is explicitly named in the course syllabus; Rand’s nonlinear-vibration notes provide the perturbation context.^[2,7]

The approximation is local. It must not be extrapolated through the saddle or used as a polynomial substitute for an entire quarter turn. As a separatrix is approached, the conservative period grows strongly; the global energy calculation captures behavior that a constant-frequency linear model misses.

14. A genuine unforced Poincaré return map

Choose the transverse section $\Sigma = \{\theta = 0, \omega > 0\}$. Start at $(0, v_n)$ and integrate until the next *positive-direction* zero crossing, one complete oscillation later. Define

$$v_{n+1} = P(v_n).$$

This is an event-defined return map. Plotting arbitrary successive animation frames would not produce the same object.

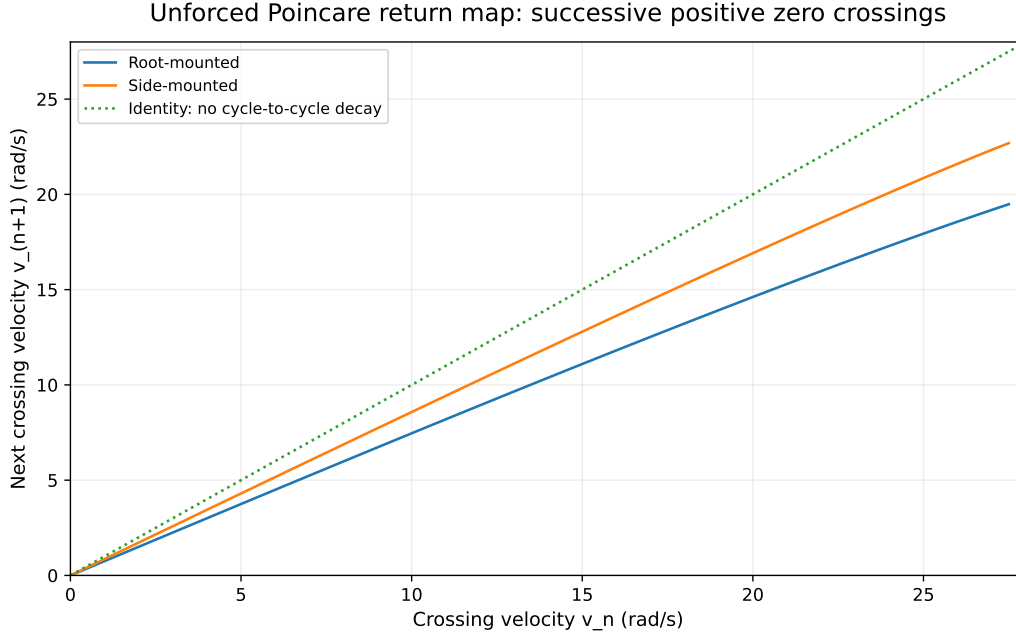


Figure 17: Next crossing speed versus current crossing speed. Both curves lie below the identity line, demonstrating cycle-to-cycle loss. The sampled range is within the sufficient aligned-well capture speeds.

For the linearized underdamped oscillator,

$$\lim_{v \rightarrow 0^+} \frac{P(v)}{v} = \exp \left[-\frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \right].$$

The limits are approximately **0.751** for root and **0.861** for side. The numerically computed small-crossing values agree. The origin is a limiting fixed point of the extended map; an equilibrium itself is not a transverse crossing.

This comparison is *per oscillation*. Root has fewer small-amplitude cycles per second, so a smaller return-map multiplier must not be mistaken for an equally large advantage per unit time. The physical-time decay rates were reported separately.

Poincaré maps and their numerical construction are part of the supplied syllabus.^[2] Here they summarize transient damping; they do not reveal a chaotic free attractor, which is incompatible with the energy structure of this model.

15. Added periodic torque: stroboscopic maps

For this separate experiment, add the same torque to both layouts:

$$\tau(t) = 0.002 \cos(2\pi 30t) \text{ N m}, \quad T = 1/30 \text{ s}.$$

This is a selected controlled excitation, not a measured human-turn waveform. A stroboscopic map samples (θ, ω) at a fixed drive phase once per T .

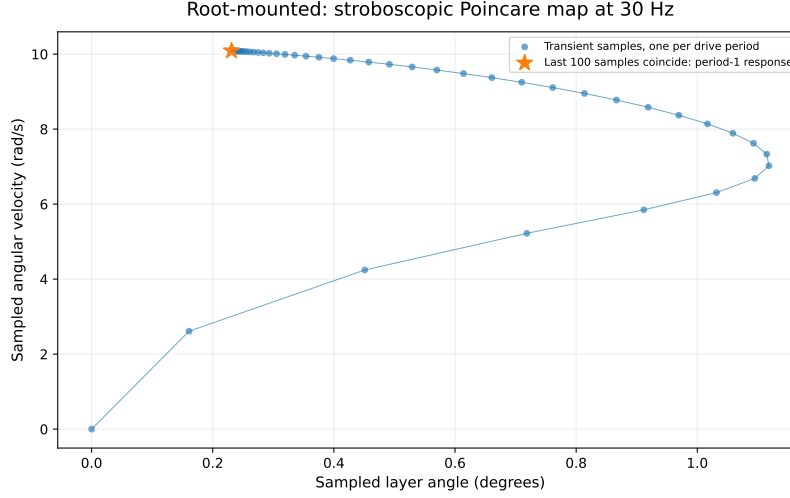


Figure 18: Root: transient samples approach one period-one fixed point. The connected transient path is only a guide between discrete samples, not an orbit in the autonomous two-dimensional state plane.

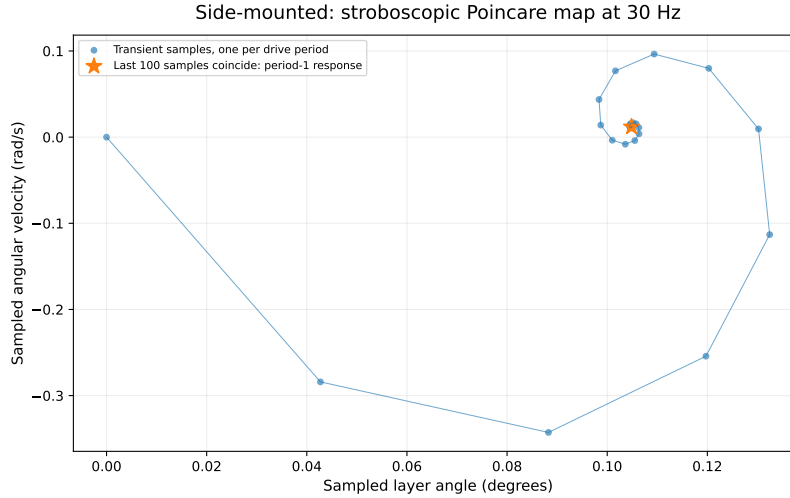


Figure 19: Side: the same forcing also reaches a period-one response. Plot scales differ to show its much smaller response; compare the numerical amplitudes, not the apparent diagram sizes.

After 260 periods, the last sampled points coalesce. Shooting refines the period-one orbit. Its peak angular magnitudes are 3.016° (root) and 0.105° (side). The forcing is near the root model's resonance, so this large contrast must not be promoted into a frequency-independent ranking.

16. Floquet stability of the forced responses

About a period- T response $\theta_p(t)$, integrate the perturbation equations

$$\dot{\boldsymbol{\eta}} = \mathbf{A}(t)\boldsymbol{\eta}, \quad \mathbf{A}(t) = \begin{bmatrix} 0 & 1 \\ -U''(\theta_p(t))/J & -c/J \end{bmatrix}, \quad \dot{\boldsymbol{\Phi}} = \mathbf{A}\boldsymbol{\Phi}, \quad \boldsymbol{\Phi}(0) = \mathbf{I}.$$

The eigenvalues of the monodromy matrix $\boldsymbol{\Phi}(T)$ are the Floquet multipliers of the stroboscopic map.^[7]

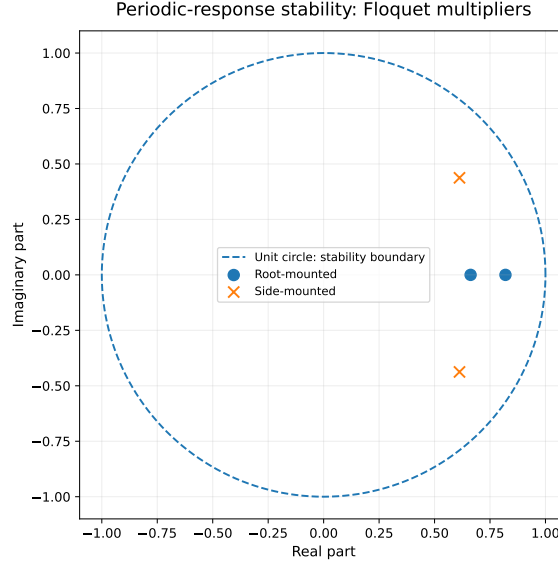


Figure 20: Both calculated multiplier sets lie strictly inside the unit circle for the selected 2 mN m, 30 Hz excitation. This establishes local attraction of the computed periodic responses.

Quantity	Root	Side
Multipliers	0.6624, 0.8196	$0.6125 \pm 0.4375i$
Largest magnitude	0.8196	0.7527
$\det \boldsymbol{\Phi}(T)$	0.542967	0.566517

Since $\text{tr } \mathbf{A} = -c/J$ is constant, Liouville's identity supplies the independent check

$$\det \boldsymbol{\Phi}(T) = \exp(-cT/J).$$

The computed determinants agree. Shooting closure errors are below 1.1×10^{-11} in the stored state norm, and halving the RK4 time step changes the final root angle by less than 10^{-9} rad.

These are local stability results for *these* forced orbits. They do not rule out other attractors at other forcing amplitudes, frequencies or initial states. This forcing is additive; no Mathieu or parametric-excitation model has been silently introduced.

17. Forced frequency response: a frequency-dependent tradeoff

The same sinusoidal torque amplitude, 2 mN m, is applied from 10 to 80 Hz in 1 Hz increments. Each point is integrated for 250 periods; the last 20 periods supply half the peak-to-peak angle. Increasing and decreasing sweeps each start from rest at their first endpoint and then continue the preceding final state.

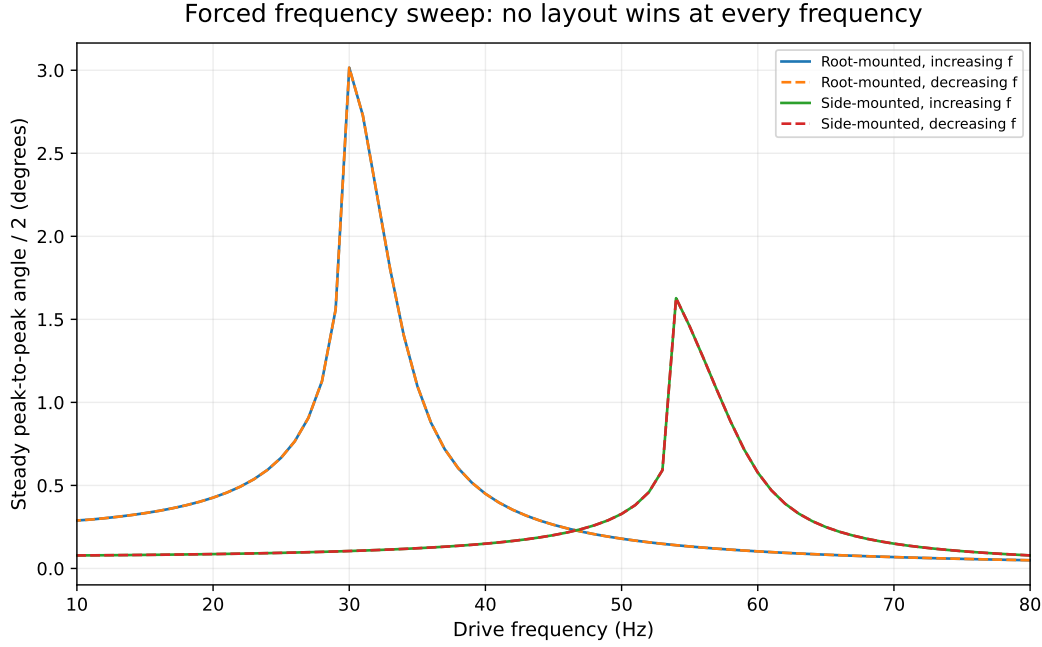


Figure 21: Sampled response branches, not a complete bifurcation diagram. Upward and downward continuations overlap at the tested frequencies and nominal forcing amplitude.

The largest sampled root amplitude is about 3.016° at 30 Hz. The largest sampled side amplitude is about 1.628° at 54 Hz. At 55 Hz, root is approximately 0.132° while side is 1.458° : the relative response reverses from the 30 Hz example.

The leftward shift relative to the small-amplitude natural frequencies is consistent with softening. At the sampled values, the last-period stroboscopic spreads are numerically tiny and the two continuation directions agree. This run does **not** demonstrate hysteresis, period doubling, quasiperiodicity or chaos. Nor does it prove those cannot occur under different forcing.

What averaging would add

For a local cubic approximation, a first-harmonic balance gives

$$A^2 \left[\left(k + \frac{3}{4} \alpha A^2 - J \Omega^2 \right)^2 + (c \Omega)^2 \right] = F^2.$$

This relation explains how amplitude-dependent stiffness bends a resonance curve. It is included as a local analytical interpretation, not as the source of the displayed numerical curve or a substitute for solving the full potential. Multiple-scales or averaging can refine such approximations, but are not needed to claim every course tool has been applied.

18. Intended and other magnetic interactions

At the reference alignment, four of the 560 changing pairs are designated intended edge–corner partners. The others remain in the calculation with their full physical contribution. Pair labels are fixed only for this diagnostic; after a quarter turn, different partners may be intended.

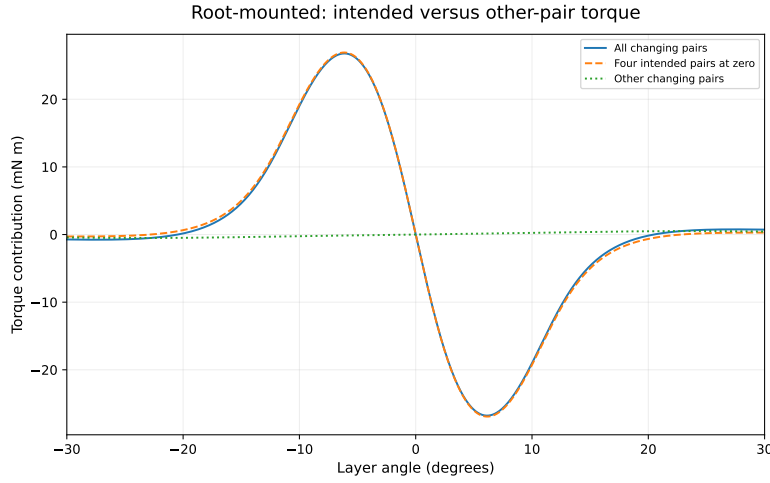


Figure 22: Root torque decomposition about the reference alignment. The full curve is the sum of the reference intended-pair and other-pair curves.

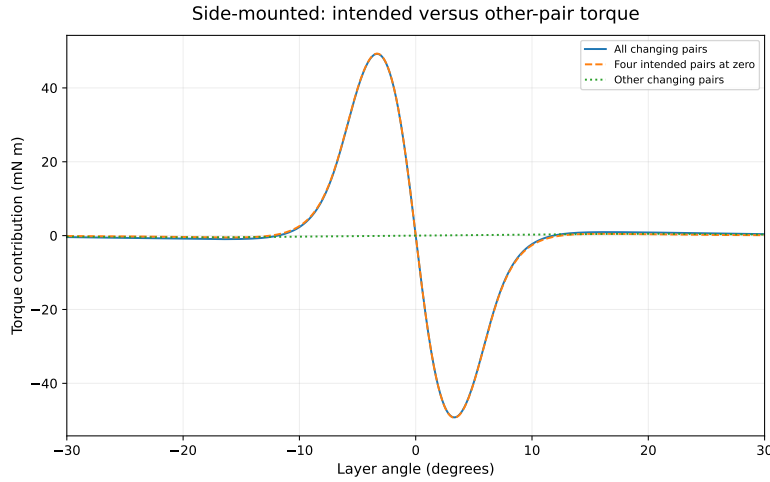


Figure 23: Side torque decomposition under the same convention. An apparently small total other-pair torque can involve cancellation, not universally weak pair fields.

The net other-pair contribution to local stiffness is approximately $-0.001282 \text{ N m/rad}$ for root and $-0.001326 \text{ N m/rad}$ for side, or -0.291% and -0.088% of the total, respectively. These small local fractions apply to the selected perfectly symmetric arrangement only. They do not establish tolerance robustness, negligible individual parasitic forces, or stability against independent piece wobble. The full-pair potential is retained when assessing the halfway wells and all other dynamics.

19. Sensitivities that separate competing mechanisms

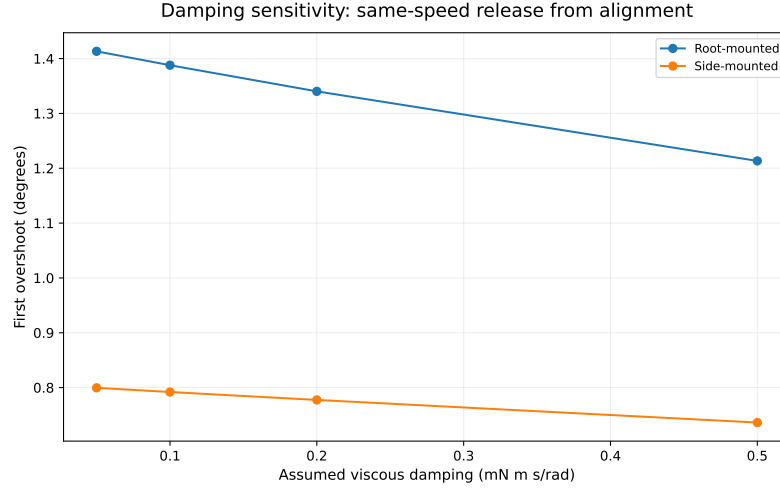


Figure 24: Same-speed release at alignment, with matched damping in both layouts. The damping values are exploratory assumptions, not fitted measurements or confidence intervals.

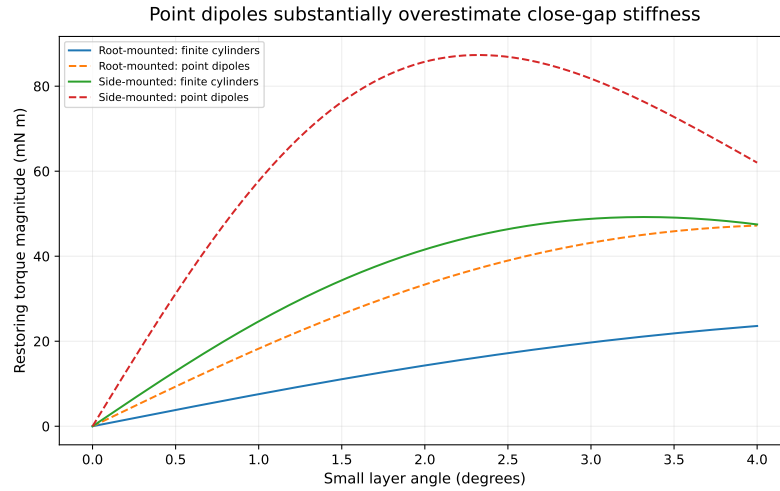


Figure 25: Finite-cylinder versus point-dipole predictions near alignment. The finite-size model is essential at this gap; the dipole stiffness is about 2.44 times the finite-cylinder stiffness.

An inertia-only diagnostic holds the root potential and nominal c fixed. Giving it the larger side inertia increases its first same-speed peak from 1.340° to 1.394° . Thus the lower root inertia *does* help when the restoring law is held fixed; the complete layout comparison is dominated by the accompanying change in magnetic stiffness.

Conversely, using the lower root inertia with the unchanged side potential decreases the same-speed peak from approximately 0.777° to 0.748° . These are diagnostic virtual systems, not additional built prototypes. The primary comparison always uses each layout's own inertia and full magnetic energy.

20. Numerical verification and untested physics

Check	Result	What it establishes
Higher magnetic quadrature	Max sampled energy difference about 2.9×10^{-10} J	Convergence of finite-source integration at checked angles
Independent double-surface integration	Relative differences below 4.4×10^{-9} for checked pairs	Agreement of two integrations of the same physical kernel
Unreduced 560-pair check	Differences near 10^{-18} J	Validity of C4 reduction for this symmetric registry
0.2° versus 0.1° field grid	Local k differences 0.029% root, 0.103% side	Angular interpolation sensitivity
RK4 step halving	Max angle differences below 2.8×10^{-9} rad in checked trace	Temporal convergence of the checked release
DOP853 versus refined RK4	Max angle differences below 1.8×10^{-10} rad	Agreement of two ODE integrations
Energy balance	Relative residual below 3.7×10^{-8}	Consistent dynamics and dissipation accounting
Capture-map step halving	0 category changes at 837 checked states	Tested map classifications are step-insensitive
Magnet-only nonoverlap	Conservative continuous separation lower bounds 0.656/0.675 mm	Cylinder separation under imposed rotation only

The separation check uses sampled separating-axis margins and a bound on intervening rotational displacement. It does not test plastic covers, tracks, screws, fit or manufacturability. Numerical convergence quantifies solver behavior, not the credibility of selected hardware parameters.

Important exclusions

No experimental torque–angle curve or ringdown has calibrated the geometry, remanence, mass proxies or damping. Mechanical preload, friction, looseness, root contact, axial motion, independent piece motion and ferromagnetic hardware response are omitted. No placement-error or polarity-error tolerance ensemble was run. The near-identical net local parasitic stiffness contributions must not be generalized beyond this symmetric model.

Course coverage and the correct next model

Phase-space dynamics, equilibrium classification, linearization, Lindstedt–Poincaré, nonlinear maps and Floquet theory are used directly.^[2] Internal resonance, nonlinear localization and energy transfer require additional coupled coordinates; they are not demonstrated by the present 1-DOF model. The next mechanically meaningful extension is one observed piece-rocking or relative-layer coordinate, with mass and contact geometry determining its kinetic energy and admissible motion. Simply adding arbitrary face angles would not supply that physics.

Defensible conclusion. The root layout improves some capture and decay metrics in this assumed system, while the side layout improves the matched first-overshoot metrics. Both have unwanted halfway minima. This is an informative model comparison, not proof of universal superiority of either physical design.

Sources and reproduction

[1] **Patent source supplied by the user.** Jerich Jaewon Lee, US10583351B2, *Puzzle cube and a method for making same*, March 10, 2020. Figures 13–15 (PDF pages 12–14); description of magnetic insertions (PDF page 20) and opposing-polarity coupling (PDF page 21). Used for arrangement and terminology, not for assumed dimensions.

<https://patents.google.com/patent/US10583351B2/en>

[2] **Course source supplied by the user.** Alexander F. Vakakis, *TAM 416 / Aero 452 Introduction to Nonlinear Dynamics and Vibrations*, Fall 2026 syllabus, page 1. Names the phase-space, perturbation, resonance, Floquet and Poincaré topics used to organize this analysis. The syllabus is not a source of the numerical model or experimental results.

[3] **Haus and Melcher, MIT.** *Electromagnetic Fields and Energy*, Section 9.3, Permanent Magnetization, especially the uniformly magnetized-cylinder example.

https://web.mit.edu/6.013_book/www/chapter9/9.3.html

[4] **Rasmus Bjørk.** *The magnetic scalar potential and demagnetization vector for a cylinder tile*, 2025. Surface-potential representation for prescribed uniform magnetization; our numerical quadrature is not the paper’s closed-form implementation.

<https://arxiv.org/abs/2501.11044>

[5] **SciPy documentation.** Complete elliptic integral (`ellipk`) and DOP853 initial-value integration (`solve_ivp`).

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.special.ellipk.html>

https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

[6] **Russ Tedrake, MIT.** *Underactuated Robotics*, Lyapunov Analysis, especially the damped-pendulum energy argument, LaSalle’s invariance principle and energy sublevel regions.

<https://underactuated.mit.edu/lyapunov.html>

[7] **Richard H. Rand, Cornell University.** *Lecture Notes on Nonlinear Vibrations*. Perturbation, nonlinear oscillators and Floquet/Poincaré background. Rand’s notes are also a recommended reference in the supplied syllabus.

<https://ecommons.cornell.edu/entities/publication/ca4770bc-fad3-4600-80ea-4c806499f525>

Reproduction and deliverables

The complete package contains the untouched baseline registry, independent finite-cylinder solver, cached data, CSV exports, all 25 figures as PNG and vector PDF, this report’s LaTeX source, and a self-contained HTML dashboard with synchronized trace-driven animation.

From the package root, install `requirements.txt`. The README gives the complete command sequence: magnetic grid, independent checks, free runs, maps, forced runs, consolidation, plots and dashboard. NumPy/SciPy/Numba/Matplotlib are sufficient; no commercial solver is required. To rerender the report, run `pdflatex analysis.tex` twice from the report directory.

Versioning caution. The scalar geometry values, explicit magnet registry and derived inertias are a frozen consistent baseline. Editing one scalar does not regenerate the registry or mass properties. After any model change, rebuild geometry, recheck symmetry and separation, recompute dynamics, and review the narrative tables. The analytic energy proof remains conditional on the resulting equilibrium structure.

Numerical source and plot data are the primary evidence for newly calculated values in this document. External references support the methods and source context, not the unmeasured quantitative performance of the prototype.